

VAPO

Value-Model-Based Reinforcement Learning
for Advanced Reasoning

60.4

AIME 2024

Surpassing DAPO by 10+ points on AIME 2024

ByteDance Seed | Base Model: Qwen2.5-32B | No SFT Data

arxiv.org/abs/2504.05118

Why Value Models?

Finer Credit Assignment for Long Chain-of-Thought Reasoning

VALUE-FREE METHODS

GRPO, DAPO

- Trajectory-level advantage
- Uniform credit across all tokens
- Monte Carlo estimates (high variance)
- No per-token signal for subtle errors



VALUE-BASED METHODS

PPO with Value Model (VAPO)

- Token-level credit assignment
- Traces each action's impact on returns
- Lower-variance estimates per token
- Generalizes across samples

Critical for multi-step reasoning where subtle step-level errors cause catastrophic failures.

Three Challenges in Value-Model-Based RL for Long-CoT

01



Value Model Bias

Long trajectories and bootstrapped learning introduce compounding estimation errors in the value model.

02



Heterogeneous Sequence Lengths

Responses vary from hundreds to thousands of tokens, requiring different bias-variance tradeoffs.

03



Reward Signal Sparsity

Only the final <eos> token receives reward; the signal must propagate across the entire chain-of-thought.

These challenges explain why value-model-free methods (GRPO, DAPO) have dominated LLM RL training.

VAPO Framework: Integrated Solutions

Each technique maps to a specific challenge it addresses

VALUE MODEL BIAS

Value-Pretraining

Pre-trains value model before RL to reduce initial bias

Decoupled-GAE

Separates GAE computation for robust value estimation

HETEROGENEOUS LENGTHS

Length-Adaptive GAE

Dynamically adjusts λ based on response length (Novel)

Token-level Loss

Normalizes loss at token level rather than sequence level

REWARD SPARSITY

Clip-Higher

Asymmetric clipping that encourages exploration

Positive Example LM Loss

Self-imitation learning on correct trajectories

Group-Sampling

Multiple responses per prompt for better baseline

Length-Adaptive GAE is the novel contribution unique to VAPO — all other techniques are adapted from prior work (DAPO, GRPO, VC-PPO).

Length-Adaptive GAE

Dynamic λ Adjustment for Variable Response Lengths

SHORT RESPONSES

$\lambda \rightarrow \text{Lower}$

- Lower variance estimates
- Tighter credit assignment
- Less noise, more precision

LONG RESPONSES

$\lambda \rightarrow \text{Higher}$

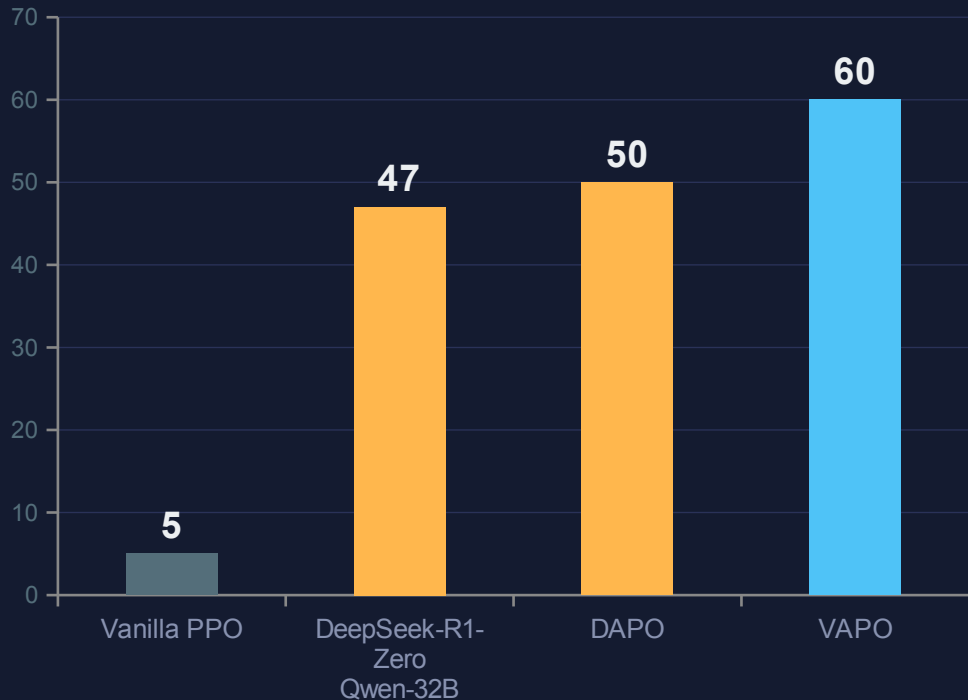
- Lower bias estimates
- Combats compounding errors
- Better for long reasoning chains



Addresses the heterogeneous sequence length problem unique to long-CoT reasoning. Contributes 15 points in ablation (45 $\rightarrow$ 60).

AIME 2024 Results: VAPO Scores 60

10+ points above DAPO and DeepSeek-R1-Zero | avg@32 | Qwen2.5-32B base, no SFT

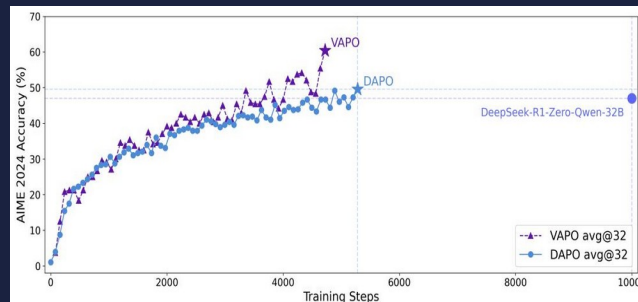


+10.4

pts over DAPO

+13.4 pts over DeepSeek-R1-Zero-Qwen-32B

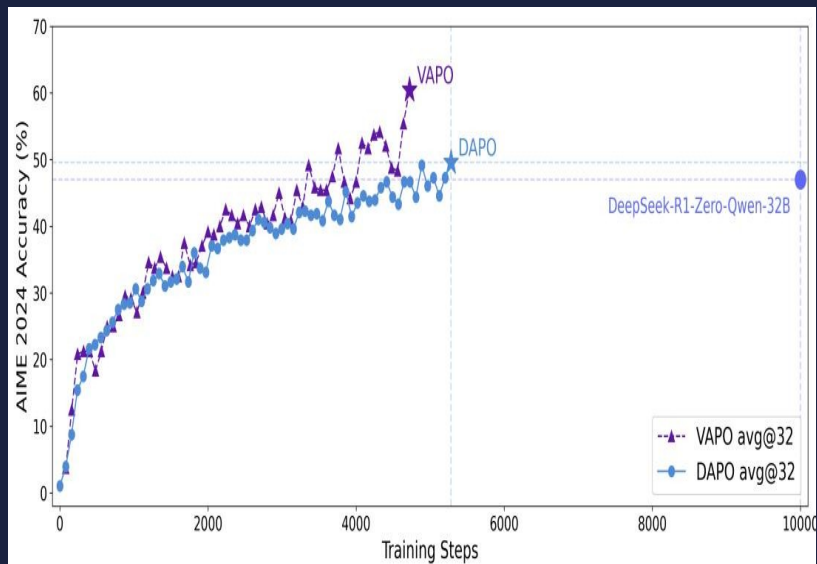
TRAINING CURVE



Training Efficiency & Stability

VAPO reaches SOTA in ~5,000 steps with zero training crashes across multiple runs

ACCURACY OVER TRAINING STEPS



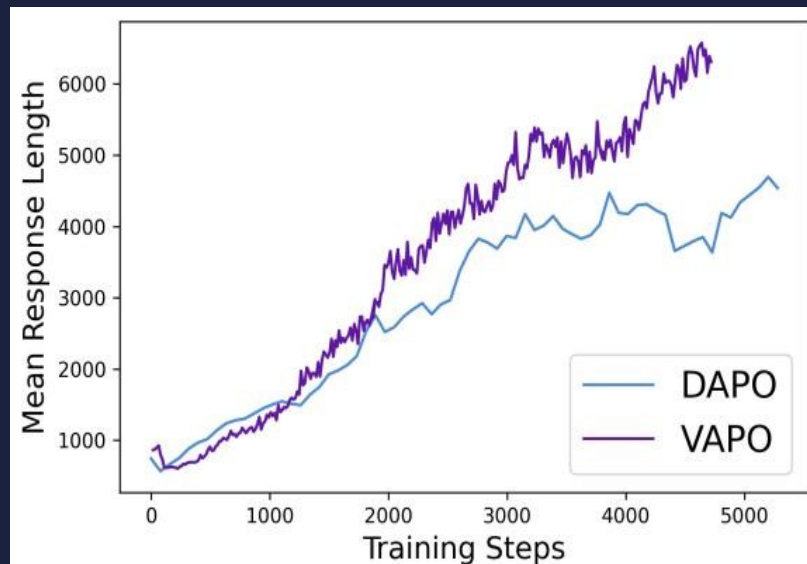
~5K

steps to SOTA

60%

accuracy @ step
4,500

MEAN RESPONSE LENGTH OVER TRAINING



0

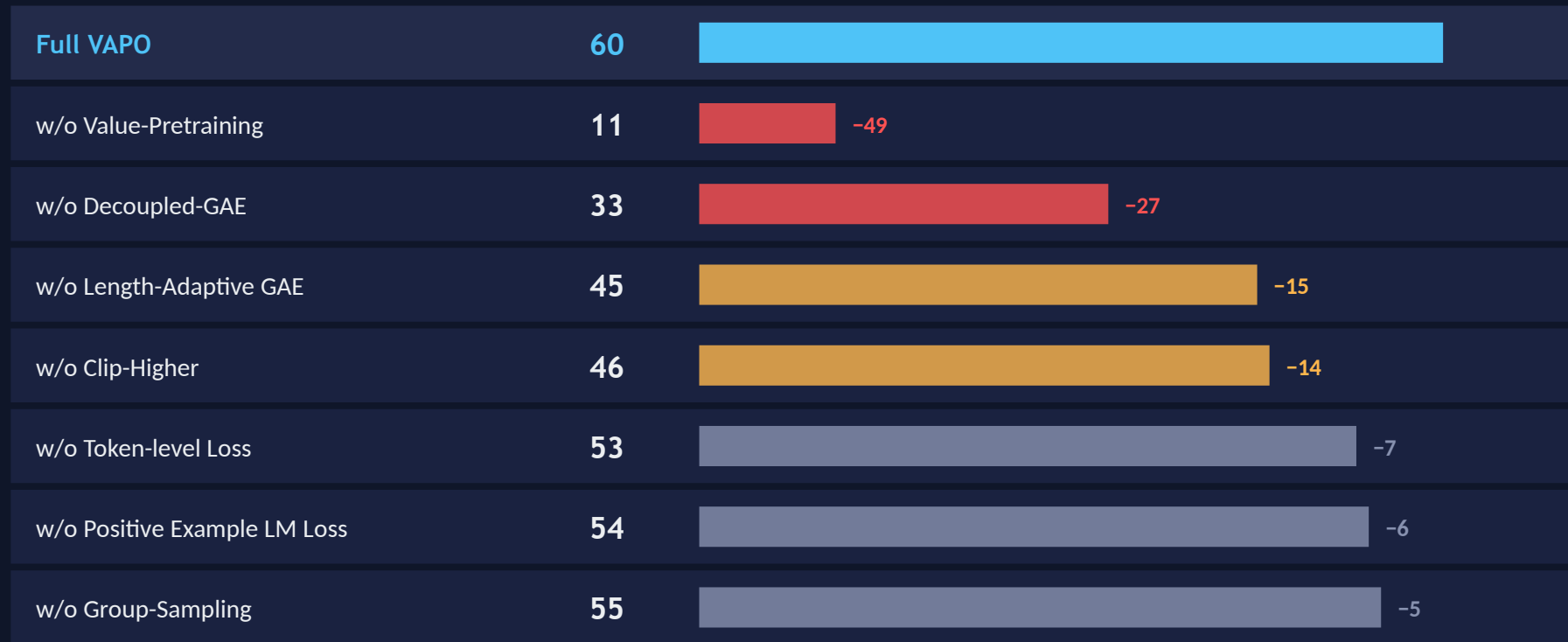
training crashes

~6K

tokens (longer
reasoning)

Ablation: Every Component Matters

Value-Pretraining alone adds 49 points (11 → 60)



Critical: Without Value-Pretraining, value-model-based RL barely functions (score: 11 vs 60).

Key Takeaways

Implications for Future Reasoning Model Development



Value-model-based RL can outperform value-free methods

VAPO achieves 60.4 on AIME 2024, surpassing DAPO (50) and DeepSeek-R1-Zero (47) — all on the same Qwen2.5-32B base model with no SFT data.



Value-Pretraining is the single most impactful component

Contributing 49 points (11 → 60), it is the prerequisite for value-model-based RL to function in long-CoT settings.



Length-Adaptive GAE solves the heterogeneous length problem

A novel technique that dynamically adjusts the bias-variance tradeoff by response length, contributing 15 points.



Superior training efficiency and stability

Reaches SOTA in ~5,000 steps with zero training crashes across multiple runs, compared to DAPO's ~5,500 steps for a lower score.



The three core challenges must be addressed together

Bias, heterogeneous lengths, and reward sparsity each require targeted solutions — removing any major component causes severe degradation.